

(Please write your Roll No. immediately)

Roll No. _____

First Term Examination (Regular)

Sixth Semester [B.Tech] (CSE,IT,EEE)

February 2018

Paper Code: ETEE-310

Subject: Microprocessor & Microcontroller

Time: 1½ Hrs

Max. Marks: 30

Note: Question No.1 is compulsory. Attempt any two questions from remaining.

- Q1. (a) Write the instructions to move value 5FH in memory location 3000H using indirect addressing mode in 8085 microprocessor. (2)
- (b) Explain the concept of segmented memory in 8086? (2)
- (c) Explain the function of ALE pin of 8085. (2)
- (d) What is wrong with MOV 1000H, BX instruction? (1)
- (e) Explain SCASB instruction with example. (2)
- (f) Explain the function of READY pin in 8085. (1)
- Q2. (a) Draw and explain the timing diagram of MVI A, 25H instruction. (5)
- (b) Explain: DB, ASSUME and EQU assembler directives with examples. (5)
- Q3. (a) Explain all the addressing modes of 8086 with suitable examples. (5)
- (b) Explain different interrupts of 8085 and their vectored locations. What is a non-maskable interrupt? (5)
- Q4. (a) Write an assembly language program to generate Fibonacci series upto first ten terms using 8085 instructions. (5)
- (b) Write an assembly language program to find square of a 16-bit number stored at location 2000H of data segment. Also store the result at location 3000H. (5)

NAME OF SUBJECT: MICROPROCESSOR AND MICROCONTROLLER
DURATION: 1:30 HRS

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NOTE: Q1 IS COMPULSORY ATTEMPT ANY TWO FROM THE REMAINING

Q. No. 1 Attempt any five

2*5

- What is transparent latch and why is it necessary to use a latch with output devices such as LEDs?
- The memory address of the last location of a 1K byte memory/chip is given as FBFFH. Specify the starting address.
- Explain why unconditional RET instruction requires 4T states to fetch whereas conditional RET instruction requires 6T states for fetch.
- Explain ASSUME and OFFSET assembler directives used in 8086
- Write a routine to transmit logic 1 through SOD line of 8085
- How Type 2 dedicated interrupt is generated. What is its vector location?

Q. No. 2

- Draw the timing diagram for 8086 minimum mode memory read cycle. 4
- Construct the binary code for the MOV CS: [BX], DL instruction. The opcode for MOV is 100010 4
- Explain the difference between LEA and MOV instruction 2

Q. No. 3

- Given that: BX = 637D, SI = 2A9B, Displacement = C237. 3
Determine the effective address (if applicable) resulting from these registers and the addressing mode:
1) Immediate (2) Direct (3) Register using BX (4) Based Indexed
- Write an ALP to produce a delay of 1 second. Consider the 8085 clocked at 5 MHz. 3
- Describe the effects on the Flags when the following instructions are executed. Before each instruction, assuming all Flags = 0 (reset) and register contents are: A = DAH; B = 06H; C = 9CH 4
MOV B, A
DCR C
ADD B
IN 22
RAL

Q. No. 4

- At a particular instant, the data in certain registers of the 8086 are as below: 2
AX=1234H; BX=3456H; CX=10 H; SI=5678H; DI=ABEFH; BP=CDEFH; CS=9087H;
DS=9035H; ES=6798H; SS=9097H. If now the instruction MOVSB is executed by the processor, state clearly what happens in the memory. Also indicate what data are left in the above registers. Assume D flag is set
- Design a microprocessor 8086 based microcomputer system having 4 chips of 4k RAM, 2chip of 2Kram, 1 input device, and 1 output device. The IO devices are interfaced in memory mapped IO. The memory map should not have any fold back addresses. Give the complete memory map of this microcomputer system. 6
- Consider the Instruction 2
2010H JUMP Disp

What is the value of Disp to go to the memory location 2005H.